
Global CO₂ Emissions

Trends, Regional Shifts, and Per Capita Inequalities

1750 – 2024 | Global Scope

For: Policy Analysts & Climate Negotiators

Sources: Global Carbon Budget 2025 · NASA Mauna Loa Observatory · Our World in Data

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

Mauna Loa Observatory, NOAA · February 2026

429

parts per million (ppm)

February 2026 · Mauna Loa Observatory

+53%
vs. pre-industrial baseline

Pre-Industrial
(~1750)

~280 ppm

Keeling Curve
begins (1958)

~315 ppm

Today
(Feb 2026)

429 ppm

Key Context

- ▶ The 429 ppm level represents more than a 50% increase over the pre-industrial baseline of ~280 ppm.
- ▶ This increase is at least 1.5× faster than the natural rise at the end of the last ice age — achieved in ~200 years vs. thousands.
- ▶ Ground-based and satellite measurements confirm sharp acceleration driven by fossil fuel combustion.

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

Annual global CO₂ from fossil fuels & industry, 1750–2024 · figure_2 reference



Trajectory Highlights

- ▶ Before the Industrial Revolution, CO₂ emissions were negligible. Growth was slow until the mid-20th century.
- ▶ Post-1950 acceleration was driven by expanded coal, oil, and gas use — primarily in industrialised nations.
- ▶ Despite recent slowing, global emissions have not yet peaked — continued growth in emerging economies offsets declines in the US and EU.

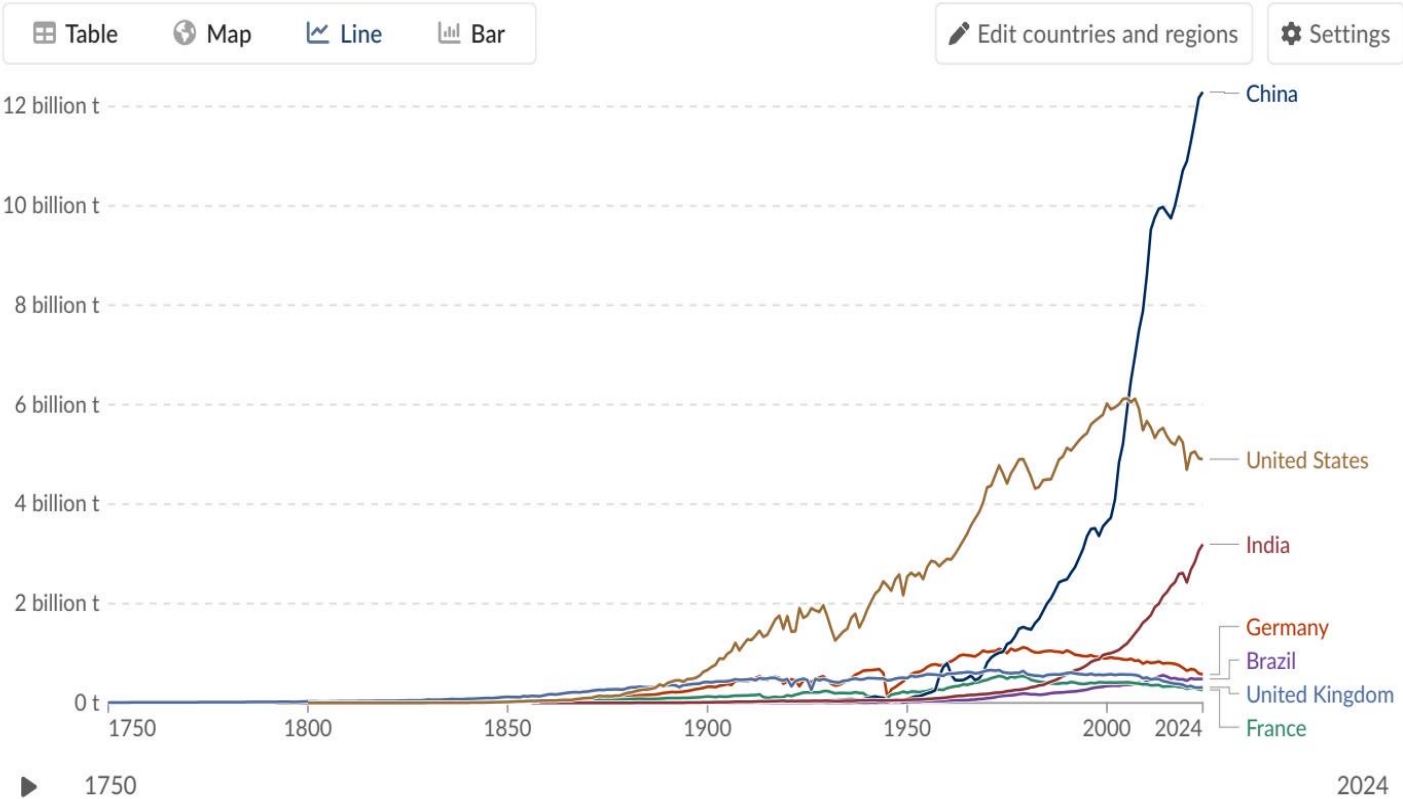
Note: figure_2 (annual totals line chart, 1750–2024) provides the full visual trajectory.

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

Annual total CO₂ by major emitter, 2024 estimates · figure_3

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.



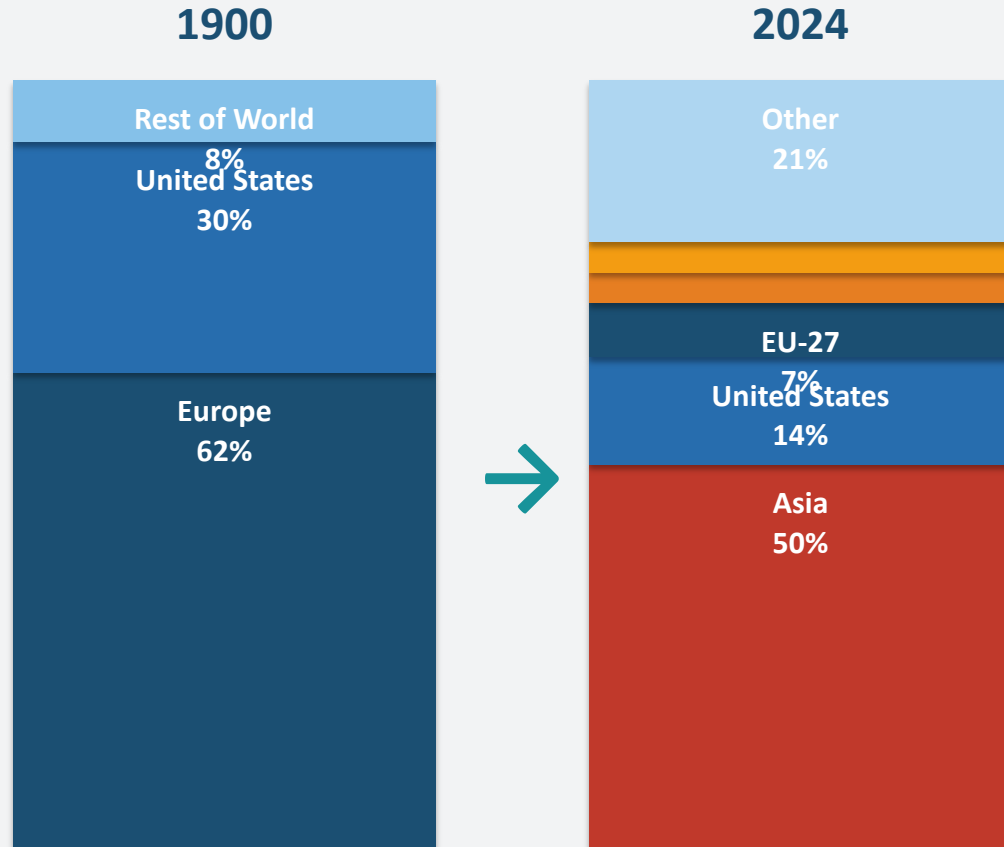
Data source: Global Carbon Budget (2025) - [Learn more about this data](#)
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

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Country / Region	CO ₂ (Bt/yr)	Global %	Trend
China	~12.5	~27%	↑ slight increase
United States	~5.0	~14%	↓ declining
India	~3.0	~8%	↑ rising fast
EU-27	~2.7	~7%	↓ declining
Germany	~0.6	~1.7%	↓ declining
Brazil	~0.5	~1.4%	→ stable
France	~0.3	~0.8%	↓ declining
United Kingdom	~0.3	~0.9%	↓ declining

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

Regional emission shares: 1900 vs. 2024 · Dramatic geographic shift

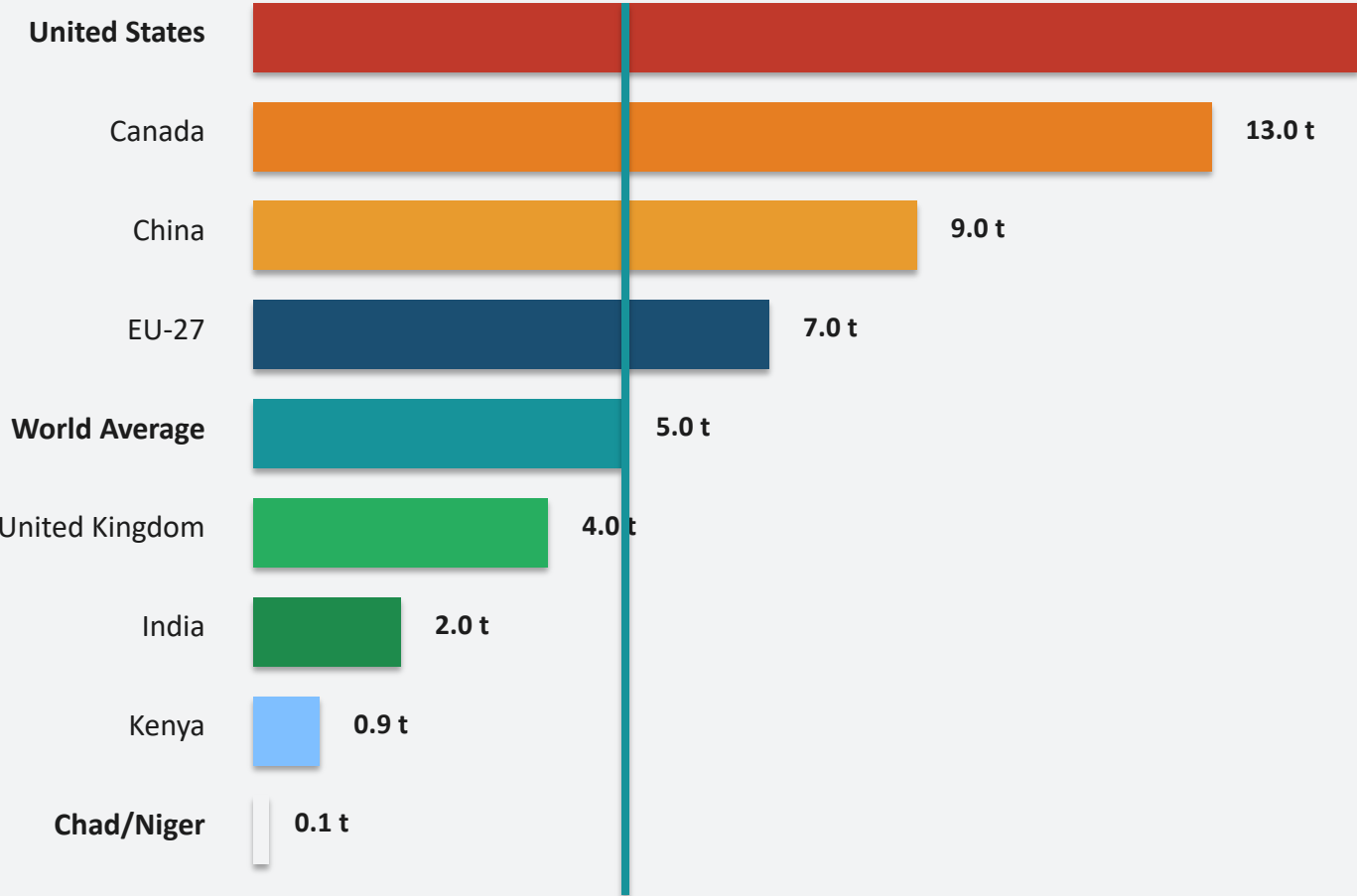


What Changed?

- ▶ In 1900, Europe and the US together accounted for more than 90% of global CO₂ emissions.
- ▶ By 1950, they still represented over 85% of emissions.
- ▶ Today, Asia — home to ~60% of the world's population — produces roughly 50% of all emissions.
- ▶ The US + Europe combined now account for less than one-third of global emissions.
- ▶ Africa and South America each contribute only 3–4%, comparable to international aviation and shipping combined.

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

Per capita CO₂ emissions (tonnes/person/year), 2024 [World figure_1](#)



The average American emits in under 2 days what a person in Chad emits in an entire year — a 150× gap.

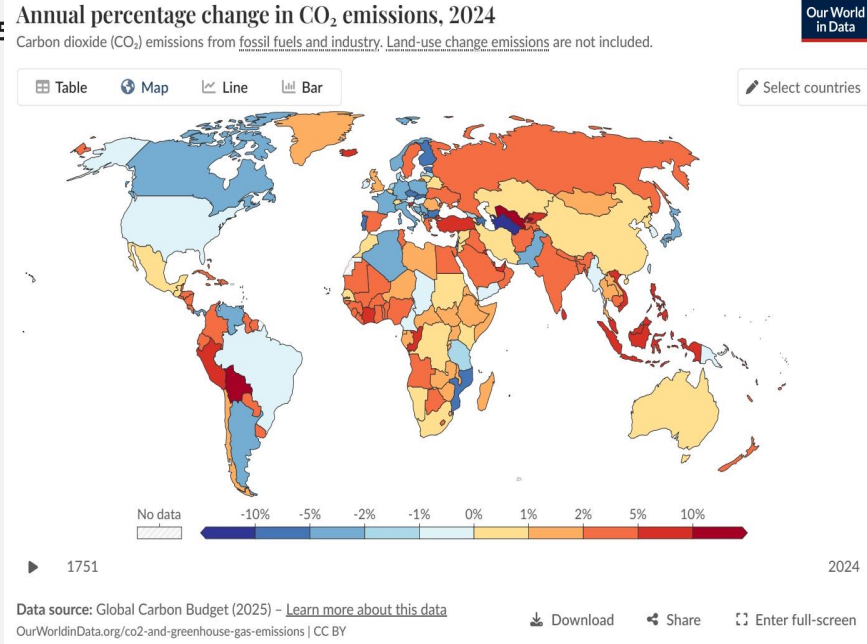


Figure 1: Per capita CO₂ over time (Our World in Data)

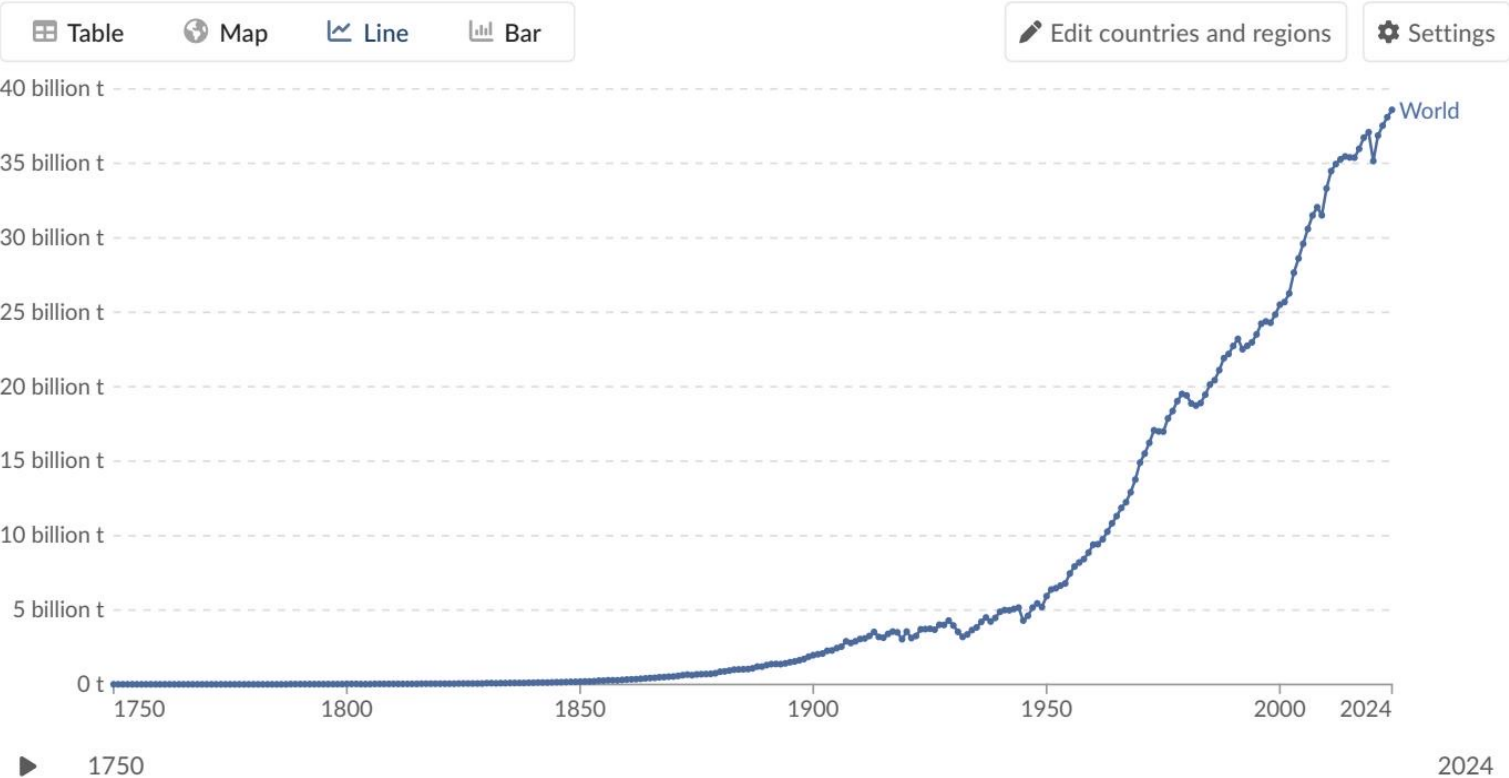
2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Annual % change in CO₂ emissions, 2024 · figure_4

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

Our World
in Data



Data source: Global Carbon Budget (2025) – [Learn more about this data](#)
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

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2024 Regional Highlights

- US & Europe (blue): –1% to –5% decline in emissions.
- Notable declines driven by renewables expansion and efficiency gains.
- Russia & SE Asia (orange/red): +2% to +10% increases.
- Several African nations also recorded increases.
- Parts of South America: mixed — some very large % increases (>10%), others declining.
- Overall: declining nations are offset by rising ones; global total still increasing.

Figure 4: Annual % change in CO₂ emissions, 2024 (Our World in Data)

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Similar prosperity levels — very different CO₂ outcomes · Policy implications

Countries with similar living standards can have very different per capita emissions depending on their energy mix.

Country	Per Capita CO ₂ (t/person/yr)	Nuclear Share	Renewables Share	Fossil Fuel Dependency	CO ₂ Outcome
France	~4 t	~70%	~25%	Low	LOW ✓
United Kingdom	~4 t	~15%	~40%	Moderate	LOW ✓
Germany	~9 t	~4% (exit)	~50%	High (coal/gas)	HIGHER ✗
Netherlands	~9 t	~3%	~35%	High (gas)	HIGHER ✗

Policy Insight

France's high nuclear share (~70%) and growing renewables keep per capita CO₂ at ~4 t/person — the same as the UK, which leads on offshore wind. Germany, which accelerated its nuclear phase-out and retains a high reliance on coal and gas, emits roughly twice as much per capita. This demonstrates that energy policy — not just prosperity or population — is the key driver of per capita CO₂ outcomes.

Key Takeaways

Emissions Are Still Rising, But the Map of Responsibility Is Shifting

1

35+ Billion Tonnes and Still Rising

Global annual CO₂ emissions from fossil fuels and industry exceed 35 Bt and have not yet peaked, despite slowing growth rates.

2

China's Dominant Share

China alone accounts for over 25% of global emissions (~12.5 Bt) — more than double the United States (~5 Bt). India (~3 Bt) is the third-largest and rising fast.

3

A 150× Per Capita Inequality

The gap between the highest per capita emitters (US, Australia ~15 t/person) and the lowest (Chad, Niger ~0.1 t/person) is approximately 150-fold — a critical dimension for climate justice negotiations.

4

Historic Shift in Responsibility

From >90% in Europe + US in 1900 to Asia producing ~50% today. The US and Europe are declining in share but their large cumulative historical emissions remain significant.

5

Policy and Energy Mix Are Decisive

Countries with similar prosperity can have very different per capita emissions (e.g., France ~4 t vs. Germany ~9 t). Nuclear and renewable energy adoption — not just wealth — determines CO₂ outcomes.